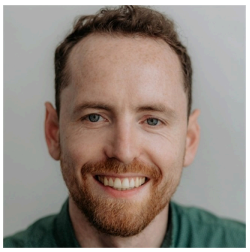
Dr. John D. Kirwan

Computational Biologist · Bioinformatics & Machine Learning · Biostatistics

La Laguna, Spain • jkirwan.org • Contact form



KEY SKILLS

Python · R · PyTorch · Machine Learning · Deep Learning · Bayesian Statistics · Protein Structure Prediction · Structural Bioinformatics · Metagenomics · Docker

PROFILES

JohnKirwan
johndkirwan
0000-0001-5537-3574
johnkirwan

LANGUAGES

English (Native)
Spanish (B2)
Swedish (B2)

PROFILE

Computational biologist and data scientist pairing rigorous statistical modelling — Bayesian inference and deep learning — with hands-on ML engineering and reproducible pipelines. Nearly four years at the interface of biotech and AI in an industry research setting, spanning biomolecular structure prediction, generative molecular design, and metagenomic analysis, and including a contribution to a NeurIPS 2025 publication on uncertainty quantification.

SELECTED HIGHLIGHTS

- Drug-discovery ML: structure prediction, docking and binding-affinity modelling in industry R&D
- Bayesian multilevel modelling (R, Stan) underpinning peer-reviewed work in quantitative biology
- Open-source R packages with browser-based Shiny apps: radiatR (on CRAN) and luxR

RECENT EXPERIENCE

Data Scientist / Bioinformatician *Freelance, La Laguna, Spain* (Jul 2026-present)

Bioinformatics Scientist *Arquimea Research Centre, La Laguna, Spain* (Jul 2022-Mar 2026)
Worked at the interface of the AI and biotech teams on structure-based drug discovery, generative molecular design, and applied machine learning — the link between the computational tools and the biology.

- Contributing author on a NeurIPS 2025 paper on uncertainty quantification for deep regression: helped develop the method in PyTorch and ran the empirical validation
- Built a modular synthetic-data-generation pipeline (Python, Biopython) orchestrating GNINA, DiffDock and AlphaFold 2, including dataset sourcing and inter-tool format conversion
- Predicted structures and binding affinities for protein and protein–peptide complexes, including metal-ion and ATP cofactors, and designed de novo peptides, using Boltz, BoltzGen, AlphaFold, Foldseek and BioEMU
- Developed the binding-affinity objective for the multi-objective small-molecule optimisation engine of Uncharted, a venture that spun out as Uncharted Chem
- Applied AI tools and data analysis to cancer-research and gut-microbiota (metagenomic) programmes; prototyped a novel operon-design approach for extremophile bacteria and ran statistical power analyses for preclinical studies
- Ran GPU workloads on an NVIDIA DGX cluster and AWS; Docker, Git and Azure DevOps within an Agile/Scrum team

Postdoctoral Researcher *Stazione Zoologica Anton Dohrn, Naples, Italy* (Oct 2019-Jul 2022)
Quantitative sensory biology within an international Human Frontier Science Program (HFSP) collaboration, with emphasis on statistical modelling and signal processing of behavioural data.

- Analysed experimental data using Bayesian multilevel regression models in R and Stan
- Designed and executed behavioural experiments with automated data acquisition pipelines
- Contributed to interdisciplinary publications spanning biology, optics, and computation

EDUCATION

PhD Biology (Vision Science) *Lund University, Lund, Sweden* (May 2013-Jun 2018) Thesis: ‘Spatial Vision in Diverse Invertebrates’.

MSc by Research Biology (Molecular Evolution) *University College Dublin, Dublin, Ireland* (Sep 2008-Sep 2010) Thesis: ‘The Molecular Evolution of Hearing in Mammals’

Dr. John D. Kirwan Bioinformatics Scientist

ADDITIONAL EXPERIENCE

Project Manager (contract) *Lund University, Lund, Sweden* (Jun 2019-Aug 2019) Short-term research management contract supervising experimental data collection and a masters student, bridging doctoral and postdoctoral positions.

Doctoral Researcher *Lund University, Lund, Sweden* (May 2013-Jun 2018) Computational and statistical analysis of visual systems in invertebrates, combining experimental biology with quantitative modelling.

- Built statistical models using Bayesian multilevel regression in R and Stan; additional analysis in Python and Matlab
- Developed optical and geometric models to estimate spatial resolution theoretically
- Designed novel experimental protocols for quantifying animal visual performance

ADDITIONAL EDUCATION

BSc (Hons) Zoology *University College Dublin, Dublin, Ireland* (Sep 2004-Jun 2008)

- First Class Honours; UCD Zoology Medal (2008)

OPEN-SOURCE SOFTWARE

- **radiatR** — CRAN R package and Shiny app for circular-arena tracking data; imports from DeepLabCut, SLEAP, EthoVision and other trackers, circular statistics and figures
- **luxR** — R package and Shiny app for modelling underwater light environments, photoreceptor excitation and visual contrast

AWARDS & CERTIFICATIONS

- Deep Learning with PyTorch — IBM — 2026
- Pharmaceutical Bioinformatics & Applied Pharmaceutical Bioinformatics — Uppsala University — 2020 Pass with Distinction
- Customising your models with TensorFlow 2 — Imperial College London — 2025
- Valohai MLOps Fundamentals Certification — Valohai — 2025
- Applied Software Engineering Fundamentals Specialization — IBM — 2023 Covers Git, GitHub, Linux, and Python, and the fundamentals of software engineering.
- Writing in the Sciences — Stanford University (online) — 2019 Statement of Accomplishment with Distinction
- Mountain Leader Award — Mountain Leader Training Scotland — 2011 Professional certification to lead hillwalking groups.

SELECTED PUBLICATIONS

Uncertainty Quantification for Deep Regression using Contextualised Normalizing Flows Adriel Sosa Marco, John D. Kirwan, Alexia Toumpa, Simos Gerasimou *The Thirty-ninth Annual Conference on Neural Information Processing Systems, 2026* www.proceedings.com/content/085/085713-1692open.pdf

Phenotypic response to food availability in sea urchin larvae and impact of light during development and growth Maria Cocurullo, John D. Kirwan, Maria Ina Arnone *Frontiers in Ecology and Evolution, 2025* www.frontiersin.org/journals/ecology-and-evolution/articles/10.3389/fevo.2025.1548208/full

A Model of Decentralized Vision in the Sea Urchin Diadema africanum Tianshu Li, John D. Kirwan, Maria Ina Arnone, Dan-Eric Nilsson, Giancarlo La Camera *iScience, 2023* www.cell.com/iscience/fulltext/S2589-0042(23)00372-3